

**Synligare utveckling av inbyggda fordonssystem –  
visualiserad kravhantering och samverkan**

**Visibler Development of Embedded Automotive Systems -  
Visualised Requirements Management and Collaboration**

|                            |                           |
|----------------------------|---------------------------|
| <b>Report type</b>         | <b>Deliverable D3.2</b>   |
| <b>Report name</b>         | <b>Tooling Prototypes</b> |
| <b>Dissemination level</b> | <b>Public</b>             |
| <b>Status</b>              | <b>Initial Release</b>    |
| <b>Version number</b>      | <b>0.5</b>                |
| <b>Date of preparation</b> | <b>2014-06-30</b>         |

---



**Authors**

---

**Editor**

Jan Söderberg

**E-mail**

jan.soderberg@systemite.se

**Authors**

Jan Söderberg

**E-mail**

jan.soderberg@systemite.se

Henrik Lönn

henrik.lonn@volvo.com

**The Consortium**

---

Volvo Technology Corporation AB - Autoliv AB - Arccore AB- Systemite AB - Semcon Sweden AB



**Revision chart and history log**

---

| <b>Version</b> | <b>Date</b> | <b>Reason</b>     |
|----------------|-------------|-------------------|
| 0.1            | 2014-01-01  | Document skeleton |
| 0.5            | 2014-06-30  | Initial Release   |



**Table of contents**

|                                               |    |
|-----------------------------------------------|----|
| Authors.....                                  | 3  |
| Revision chart and history log.....           | 5  |
| Table of contents .....                       | 7  |
| 1 Introduction .....                          | 9  |
| 2 EATOP platform and plugins .....            | 10 |
| 2.1 EATOP .....                               | 10 |
| 2.1.1 Description .....                       | 10 |
| 2.1.2 Installation instructions.....          | 10 |
| 2.2 Error Model Synthesis.....                | 10 |
| 2.2.1 Description .....                       | 10 |
| 2.2.2 Installation instructions.....          | 10 |
| 3 SystemWeaver .....                          | 11 |
| 3.1.1 Description .....                       | 11 |
| 3.1.2 Installation instructions.....          | 11 |
| 3.2 Modelling Support .....                   | 11 |
| 3.3 EAST-ADL Exchange .....                   | 12 |
| 3.4 Views.....                                | 12 |
| 3.5 Metrics .....                             | 12 |
| 4 EnterpriseArchitect EAST-ADL Exchange ..... | 13 |
| 4.1.1 Description .....                       | 13 |
| 4.1.2 Installation instructions.....          | 13 |
| 5 Summary.....                                | 14 |
| 6 References.....                             | 15 |





**1 Introduction**

---

This document is the report on results of work package 3 (WP3) of Project Synligare.

The overall effort in Project Synligare is to address specification of complex automotive embedded systems to provide an overview, ease review and management of requirements by means of defining structured information models for representing system specification, which will enable automatic generation of views and metrics. The purpose is to improve collaborative development of automotive embedded systems in comparison with the typical situation in which there is no common specification language. This involves addressing challenges like relating requirements and design with respect to configurations in a product family by applying various relations and searches in structured requirement information based on existing specification languages like EAST-ADL. Further, it involves management of specifications and requirements in accordance with ISO 26262. The aim is that the outcomes of Project Synligare should be tried on an example system, to validate the structured information models, the views and the metrics.

This chapter introduces WP3 by defining purpose, scope and method.

---

## **2 EATOP platform and plugins**

---

---

### **2.1 EATOP**

---

The EAST-ADL Tool Platform, EATOP, provides a reference implementation of EAST-ADL where tool plugins with various purpose can be plugged in. It is provided as an open source Eclipse project.

---

#### **2.1.1 Description**

---

This section refers to the platform part of EATOP, which is the basis for further tool plugins

---

#### **2.1.2 Installation instructions**

---

EATOP can be downloaded from [www.eclipse.org/eatop](http://www.eclipse.org/eatop). It is unzipped to any local directory and executed from there.

---

### **2.2 Error Model Synthesis**

---

The Error Model Synthesis plugin supports analyses related to dependability.

---

#### **2.2.1 Description**

---

Error propagation modeling is the basis for failure modes and effects analysis (FMEA) and fault tree analysis (FTA). This plugin automatically generates the error propagation model from a nominal model, supports model manipulations and is able to invoke an analysis engine (HiP-HOPS).

---

#### **2.2.2 Installation instructions**

---

The plugin is copied to the add-ins directory of the EATOP installation.

---

## 3 SystemWeaver

---

---

### 3.1.1 Description

---

SystemWeaver is a commercial tool developed and marketed by Systemite AB.

SystemWeaver is in use at various automotive companies, including Volvo Car Corporation and AB Volvo.

More information on SystemWeaver can be found at [www.systemite.se](http://www.systemite.se).

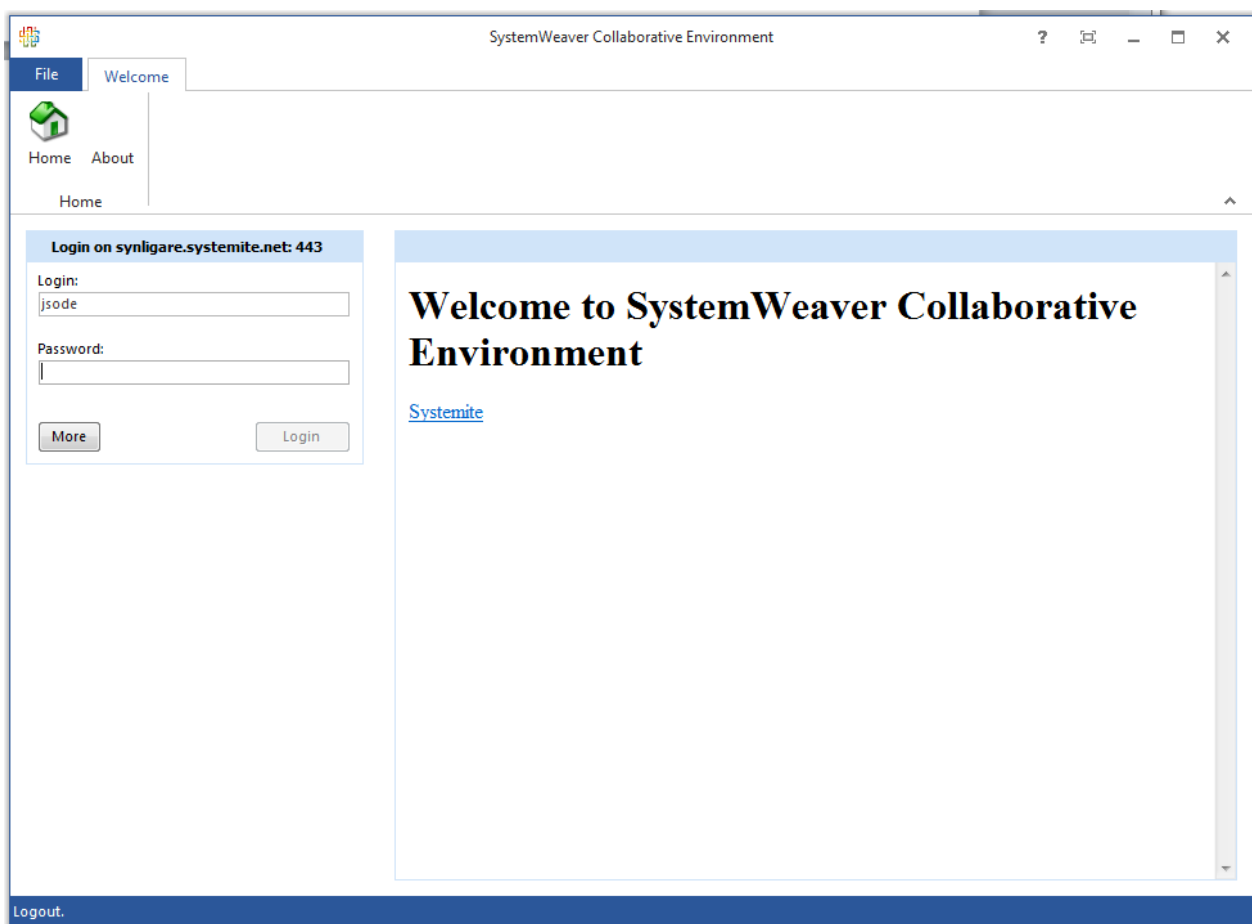
---

### 3.1.2 Installation instructions

---

The SystemWeaver tool client can be installed from <http://apps.systemite.se/Synligare/setup.exe>.

The address to be used for the login is [synligare.systemite.net](http://synligare.systemite.net), with port = 443:



Login user names and passwords for all project members have been distributed by email.

---

## 3.2 Modelling Support

---

The modeling support includes the EAST-ADL core meta model and the Requirements and Dependability extension packages.

In the Synligare project the support for EAST-ADL has been extended, and now supports allocation of requirements to DesignFunctionPrototypes.

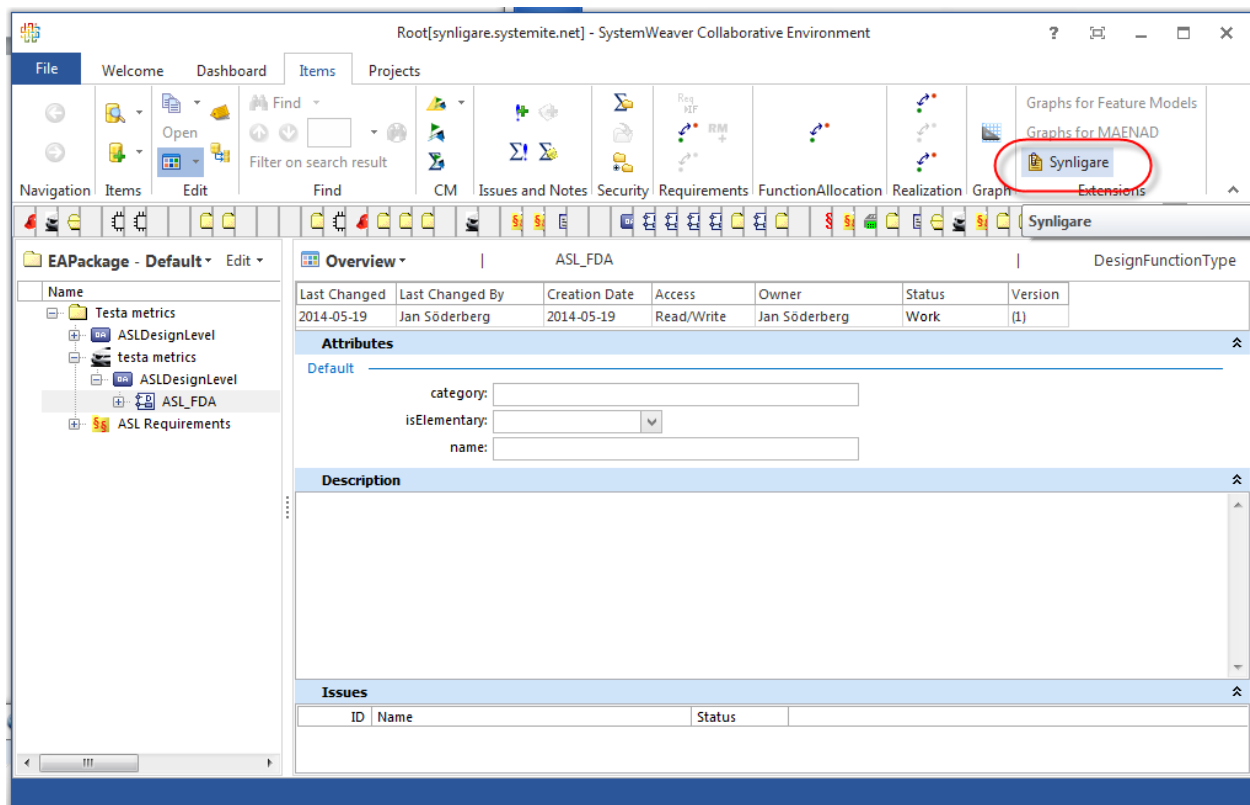
### 3.3 EAST-ADL Exchange

The EAST-ADL Exchange is based on transformation between the EAXML format and the native SystemWeaver XML format. The SystemWeaver XML format can then be used for various synchronization and consolidation operations towards a SystemWeaver database, including support for incremental imports and versioning.

### 3.4 Views

### 3.5 Metrics

The Metrics plug-in is available in the "ribbon" in the client tool application, and is available whenever an EAPackage has been selected.



---

**4 EnterpriseArchitect EAST-ADL Exchange**

---

The Enterprise Architect UML tool is a widely used, low cost UML editor. With an EAST-ADL profile, it can also be used as an EAST-ADL editor.

---

**4.1.1 Description**

---

The Enterprise Architect EAST-ADL Exchange tool allows EAXML files to be imported and exported to Enterprise Architect.

---

**4.1.2 Installation instructions**

---

The Enterprise Architect EAST-ADL Exchange tool is copied to a local directory and run as a Windows application.

Trial versions and licenses for the Enterprise Architect tool can be acquired from [www.sparxsystems.eu](http://www.sparxsystems.eu).

**5 Summary**

---

This deliverable has described tooling prototypes used to assess and validate tooling concepts developed in the Synligare project.

**6      References**

---

- [1] EATOP Eclipse Open Source Project: EAST-ADL Tool platform.  
<http://www.eclipse.org/eatop>
- [2] Sparx Systems Inc: Enterprise Architect. [www.sparxsystems.com](http://www.sparxsystems.com)
- [3] Synligare Consortium: Synligare Deliverable D1.1 Needs Identification.  
<http://www.synligare.eu/>
- [4] Synligare Consortium: Synligare Deliverable D2.1 Modeling Concepts and Methods.  
<http://www.synligare.eu/>
- [5] Synligare Consortium: Synligare Deliverable D3.1 Report on Tooling.  
<http://www.synligare.eu/>
- [6] Systemite AB: SystemWeaver. <http://www.systemite.se>